Python String Notes

Based on Chai String Practice Session

1. Creating Strings

```
chai = "lemmon chai"
```

Strings in Python are created using single or double quotes. Typing the variable name in the interpreter shows its value with quotes.

2. String Indexing & Slicing

Indexing

```
first_char = chai[0]   # Returns: "l"
```

- Strings are zero-indexed — the first character is at index `[0]`

Slicing

```
slice = chai[0:2]   # Returns: "le"
```

- Syntax: `string[start:end]` — end index is exclusive
- `chai[0:2]` returns characters at index 0 and 1, not 2

3. String Methods

.lower() and .upper()

```
print(chai.lower())   # lemmon chai print(chai.upper())   # LEMMON CHAI
```

- `.lower()` converts all characters to lowercase
- `.upper()` converts all characters to uppercase

.replace()

```
print(chai.replace("lemmon", "ginger"))   # ginger chai
```

- Arguments must be strings in quotes — e.g., `"lemmon"`, not lemmon
- Without quotes, Python looks for a variable, causing a NameError

💡 *Error encountered: NameError: name 'lemmon' is not defined — fixed by adding quotes*

4. String Formatting with .format()

```
chai_type = "masala" quantity = 2 order = "i ordered {} cups of {} chai"
print(order.format(quantity, chai_type)) # Output: i ordered 2 cups of masala
chai
```

- Use {} as placeholders inside a string
- `.format()` replaces {} in order with the provided arguments
- Useful for building dynamic messages with variables

5. Lists & .join()

```
chai_variety = ["lemon", "masala", "ginger"] print(", ".join(chai_variety)) # Output: lemon, masala, ginger
```

- Lists are created with square brackets: ["item1", "item2"]
- `.join()` combines list items into a single string, separated by the given string
- Syntax: `"separator".join(list)`

6. Common Errors Encountered

`IndentationError: unexpected indent`

Caused by accidental spaces before a print() statement. Python is whitespace-sensitive — never indent code at the top level unless inside a block (if, for, def, etc.).

`NameError: name 'lemmon' is not defined`

Caused by passing a variable name instead of a string. Always wrap text arguments in quotes when you mean a literal value.

7. Quick Reference

Method	Example	Output
chai[0]	`chai[0]`	`"l"`
Slicing	`chai[0:2]`	`"le"`
.lower()	`chai.lower()`	`"lemmon chai"`
.upper()	`chai.upper()`	`"LEMMON CHAI"`
.replace()	`chai.replace("lemmon","ginger")`	`"ginger chai"`
.format()	`order.format(2, "masala")`	`"i ordered 2 cups..."`
.join()	`", ".join(chai_variety)`	`"lemon, masala, ginger"`